Title

Next Generation Sequencing 2025/2026

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1 Setting The Environment

```r
# ============================================================
# Install and Call Required NameSpaces
# ============================================================

# Install BioConductor package if not installed
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

# Package Lists
cran_pkgs <- c("tidyverse", "corrplot", "ggfortify", "ggplot2")
bioc_pkgs <- c("edgeR", "EDASeq", "limma")

# Check, Install, and Call NameSpaces using CRAN
for (pkg in cran_pkgs) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  } else {
    library(pkg, character.only = TRUE)
  }
}

# Check, Install, and Call NameSpaces using BioConductor
for (pkg in bioc_pkgs) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    BiocManager::install(pkg)
    library(pkg, character.only = TRUE)
  } else {
    library(pkg, character.only = TRUE)
  }
}

results_dir <- "results/"
dir.create(results_dir, showWarnings = FALSE, recursive = TRUE)
```

2 Explarotary Analysis

```r
# ============================================================
# Load the Dataser from rds file
# ============================================================
rse <- readRDS("./opt/omicsdata/datasets/SRP029880.rds")
rse
```

```
## class: RangedSummarizedExperiment
## dim: 63856 54
## metadata(8): time_created recount3_version ... annotation recount3_url
## assays(1): raw_counts
## rownames(63856): ENSG00000278704.1 ENSG00000277400.1 ...
##   ENSG00000182484.15_PAR_Y ENSG00000227159.8_PAR_Y
## rowData names(10): source type ... havana_gene tag
## colnames(54): SRR975551 SRR975552 ... SRR975603 SRR975604
## colData names(175): rail_id external_id ...
##   recount_pred.curated.cell_line BigWigURL
```

The given dataset contains 63856 genes across 54 samples.

```r
# ============================================================
# Inspect Data More
# ============================================================
# Inspect row and column names of the data
message("Row names: ", paste(rownames(rse)[1:3], collapse = ", "),
        "\nColumn names: ", paste(colnames(rse)[1:5], collapse = ", "))
```

```
## Row names: ENSG00000278704.1, ENSG00000277400.1, ENSG00000274847.1
## Column names: SRR975551, SRR975552, SRR975553, SRR975554, SRR975555
```

```r
# Inspect Sample (colData) and Genes metadata (rowData)
message("Sample metadata: ")
```

```
## Sample metadata:
```

```r
print(colData(rse)[1:5, 1:5])
```

```
## DataFrame with 5 rows and 5 columns
##             rail_id external_id       study sra.sample_acc.x sra.experiment_acc
##           <integer> <character> <character>      <character>        <character>
## SRR975551   1354299   SRR975551   SRP029880        SRS478664          SRX347887
## SRR975552   1354331   SRR975552   SRP029880        SRS478665          SRX347888
## SRR975553   1354362   SRR975553   SRP029880        SRS478667          SRX347889
## SRR975554   1354395   SRR975554   SRP029880        SRS478666          SRX347890
## SRR975555   1354426   SRR975555   SRP029880        SRS478668          SRX347891
```

```r
message("\nGene metadata: ")
```

```
##
## Gene metadata:
```

```r
print(rowData(rse)[1:5, ])
```

```
## DataFrame with 5 rows and 10 columns
##                     source     type bp_length     phase           gene_id
##                   <factor> <factor> <numeric> <integer>       <character>
## ENSG00000278704.1  ENSEMBL     gene      2237        NA ENSG00000278704.1
## ENSG00000277400.1  ENSEMBL     gene      2179        NA ENSG00000277400.1
## ENSG00000274847.1  ENSEMBL     gene      1599        NA ENSG00000274847.1
## ENSG00000277428.1  ENSEMBL     gene       101        NA ENSG00000277428.1
## ENSG00000276256.1  ENSEMBL     gene      2195        NA ENSG00000276256.1
##                        gene_type   gene_name       level havana_gene
##                      <character> <character> <character> <character>
## ENSG00000278704.1 protein_coding  BX004987.1           3          NA
## ENSG00000277400.1 protein_coding  AC145212.2           3          NA
## ENSG00000274847.1 protein_coding  AC145212.1           3          NA
## ENSG00000277428.1       misc_RNA       Y_RNA           3          NA
## ENSG00000276256.1 protein_coding  AC011043.1           3          NA
##                            tag
##                    <character>
## ENSG00000278704.1          NA
## ENSG00000277400.1          NA
## ENSG00000274847.1          NA
## ENSG00000277428.1          NA
## ENSG00000276256.1          NA
```

The column names represent sample names and the row names represent gene IDs.

The sample metadata contains 175 variables for each 54 sample.

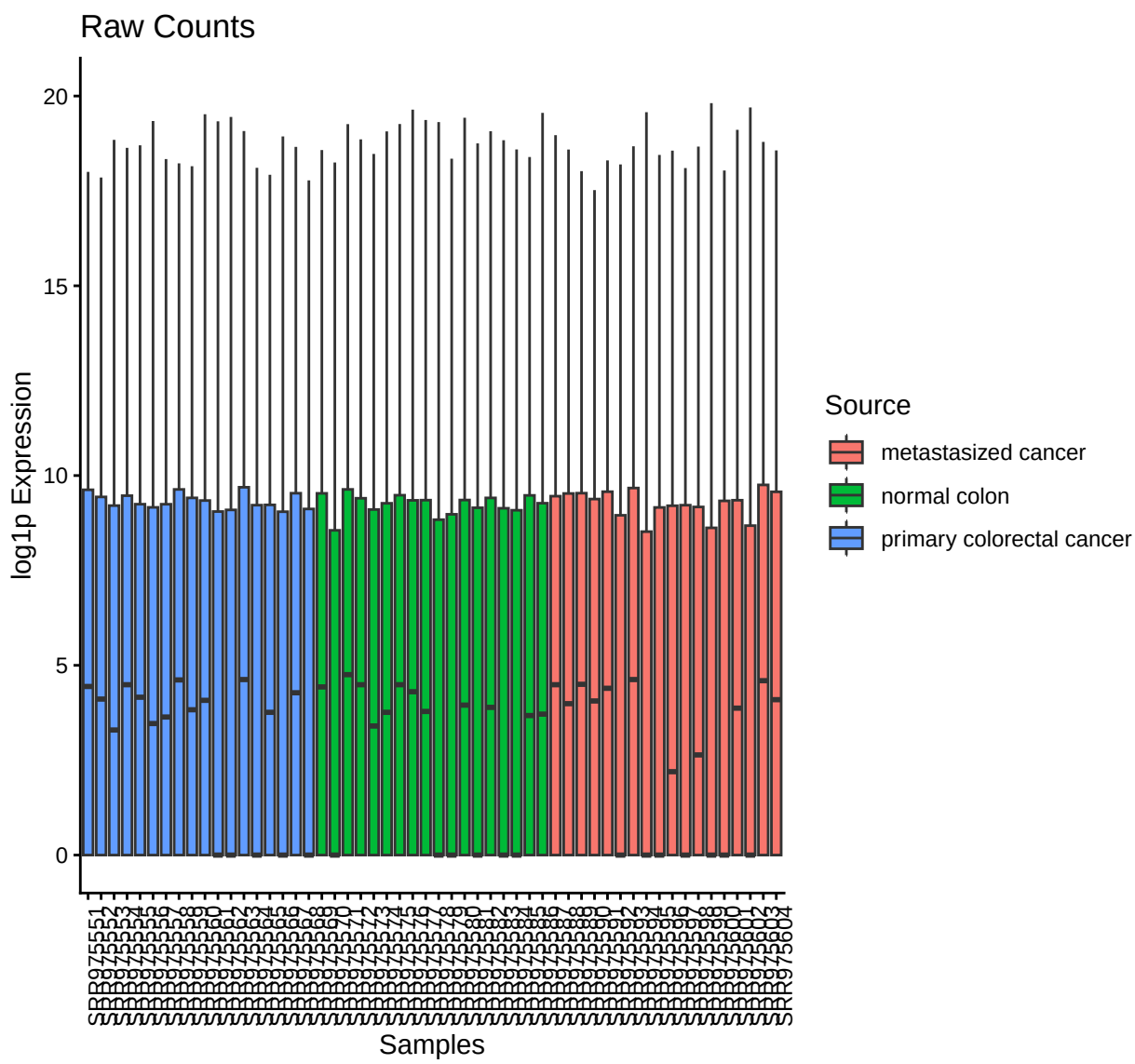
The gene metadata contains 10 variables for each 63856 genes.

```r
# ============================================================
# Extract information from attributes data to a new column
# ============================================================
rse@colData$source_name <- sapply(
  strsplit(rse@colData$sra.sample_attributes, split = "|", fixed = TRUE),
  function(x) strsplit(x[2], split = ";;", fixed = TRUE)[[1]][2]
)
```

```r
# ============================================================
# Exploratory Plots
# ============================================================
raw_counts  <- assay(rse, "raw_counts")
source_name <- rse$source_name
names(source_name) <- colnames(rse)
log1p_counts <- log1p(raw_counts)

df_long <- as.data.frame(log1p_counts) %>%
  mutate(genes = row.names(.)) %>%
  pivot_longer(cols = -genes, names_to = "samples", values_to = "expression")
df_long$source_name <- source_name[df_long$samples]
```

```r
ggplot(df_long, aes(x = samples, y = expression, fill = source_name)) +
  geom_boxplot() +
  scale_y_continuous(limits = c(0, 20)) +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(title = "Raw Counts", x = "Samples", y = "log1p Expression", fill = "Source")
```




```r
exp_plot_dir  <- file.path(results_dir, "exploratory")
ma_plot_dir   <- file.path(exp_plot_dir, "ma_plots")
md_plot_dir   <- file.path(exp_plot_dir, "md_plots")
dir.create(exp_plot_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(ma_plot_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(md_plot_dir, showWarnings = FALSE, recursive = TRUE)


combinations <- combn(c(1:ncol(rse)), 2)
```

```r
sample_names <- colnames(rse)

# MA Plot
begin <- proc.time()
for (i in 1:ncol(rse)) {
  filename <- paste0("MAplot_", sample_names[i], ".png")

  png(filename = file.path(ma_plot_dir, filename), width = 800, height = 600)

  limma::plotMA(log1p_counts, array = i,
                main = paste0(sample_names[i]), xlab = "A", ylab = "M")

  dev.off()
}
run_time <- proc.time() - begin
message("Time elapsed during MA Plotting: ", run_time[3], " seconds.")
```

```
## Time elapsed during MA Plotting: 11.91 seconds.
```

```r
# MD Plot
begin <- proc.time()
for (i in 1:ncol(combinations)) {
  pair <- combinations[, i]

  filename <- paste0("MDplot_", sample_names[pair[1]], "_vs_", sample_names[pair[2]], ".png")

  png(filename = file.path(md_plot_dir, filename), width = 800, height = 600)

  EDASeq::MDPlot(raw_counts, c(pair[1], pair[2]))
  title(main = paste0(sample_names[pair[1]], " vs ", sample_names[pair[2]]))

  dev.off()
}

run_time <- proc.time() - begin
message("Time elapsed during MD Plotting: ", run_time[3], " seconds.")
```

```
## Time elapsed during MD Plotting: 441.61 seconds.
```

3 Normalization

4 Differential Expression Analysis

5 Gene Enrichment Analysis